EQUIPMENT NAME:	**FIM III Booster Station 33/11kV Transformer 11kV Neutral Earthing Resistor**
EQUIPMENT NUMBER:	**F23-NR-601**

Procurement Package		"EL007" – Neutral Earthing Resistor
Equipment No.	**Rating**	**Equipment Name**
F23-NR-601	25A for 10s	FIM III Booster Station 33/11kV Transformer 11kV Neutral Earthing Resistor
Reference Documents		2000-F23-SLD-EL-40001 – FIM III Booster Station F23-SW-601 HV Switchboard Single Line Diagram
Reference Specifications		2000-F00-STS-EL-10008 – Electrical Enclosures Specification 2000-F00-STS-GE-10002 – Site Specific Information 2000-F00-STS-GE-10001 – Packing and Shipping Specification
Reference Standards		AS/NZS 2081 – Appendix C

Scope of Supply

Work Included

- Design, supply and fabrication of one (1) Neutral Earthing Resistor including but not limited to:
  - Fully assembled in 316SS enclosure for outdoor pad mounted installation
  - 30/0.05A 5P20 2.5VA Current Transformer
  - Junction box & terminals for all secondary connections
    - Instrumentation as appropriate
    - Wiring for instrumentation / control
    - Labelling
    - Lifting lugs, as applicable
    - Any other items which are normally supplied with this equipment
- Installation of free-issued items, including:
  - SE-330AU earthing resistor monitoring relay
  - SE-CS30 current transformer
- All ancillary items and special tools normally supplied with, or required to adjust and maintain all components associated with the equipment
- Laser engraved stainless steel equipment nameplate
- All fasteners, gaskets, packers, shims and any other items necessary for site assembly
- Factory Acceptance Testing at manufacturer's premises
- Supply of all necessary documentation, drawings and test documentation
- Packaging suitable for sea and road transport for delivery
- Spare parts list
- Delivery to site at Western Australia.

Work Excluded

- Hold-down bolts, nuts and washers at concrete foundations/structural steel
- Spare parts
- Offloading of equipment at site
- Installation and commissioning at site

Description	Unit	Duty	Supplier	Notes	Rev
General					
Equipment Number and Name Plate		Yes, on stainless steel plate.		1	A
Make		Fortress Resistors		3	A
Model				2	A
Country of Manufacture				2	A
Compliance with Australian Standards	Comply /Do Not Comply	AS 2081		3	A
Environmental Conditions					
Plant Location		Western Australia		1	A
Service Location		Outdoor, direct sunlight and rain, polluted atmosphere		1	A
Site Data		Refer to 2000-F00-STS-GE-10002 – Site Specific Information		1	A
Physical Construction					
Enclosure Material		Stainless Steel 316		1	A
Mounting		Pad Mount		1	A
Junction Box IP Rating		IP55		1	A
Resistor enclosure IP Rating		IP55		1	A
Lifting Lugs	Yes/No	Yes		1	A
Painting Specification		No surface coating		1	A
Paint Colour		N/A		1	A

Description	Unit	Duty	Supplier	Notes	Rev
Temperature Ratings					
Rated Maximum Ambient Temperature	ºC	50		1	A
Temperature Rise	ºC			2	A
Temperature Coefficient	%/ºC			2	A
Cooling		Air Natural		3	A
Doors		Bolted removable with SS handles		3	A
Electrical Supply					
System Voltage	kV	11kV ± 10%, 1ph		1	A
Phase to Earth Voltage	kV	6.35kV ± 10%		1	A
Initial Current	A	25		1	A
Duration	S	10		1	A
Frequency	Hz	50 ± 2%		1	A
Resistance	Ω			2	B
Cable Connections					
Termination Details		HV bushing box with copper busbar		3	A
Cable Boxes					
Gland Plate Materials		Aluminium 6mm		3	A
Cable Entry		Bottom Entry		1	A
Degree of Protection		IP55		1	A
Covers		Bolted removable with SS handles		3	A
Neutral Termination		Copper busbar		3	A
Earth Termination		Copper busbar		3	A
Control Circuit					
Marshalling Box / Control Panel		Required		1	A
Instrumentation Connections		Screw Cage Terminals WDU2.5 or equivalent		3	A
Test Requirements					
Factory Acceptance Test		Yes, witnessed with test values included in report		1	A
Dimension Details					
Weight	kg			2	A
Dimensions	mm	L x W x H		2	A
Shipping Details					
Weight	kg			2	A
Dimensions	mm	L x W x H		2	A

Notes
1. Fixed. Vendor shall comply with this data
2. Vendor to Advise
3. Preferred, Vendor to advise and provide reasoning for difference

Vendor Drawing and Document Requirements List (VDDR)

E – Electronic Copy H – Hard Copy MDR – Manufacture's Data Report

FAT – Factory Acceptance Testing IOM – Installation and Operating Manual

VDDR			Required with Tender	Required after Award		As Built Required
			Copies	Copies	Weeks	Copies
A		**Production Schedule**				
	A01	Production Schedule / Bar Chart	1E	1E	2	
	A02	Monthly Report		Monthly		
B		**Registers and General Drawing Requirements**				
	B01	Drawing Schedule / Register (Drawings Expected)		1E	2	
	B02	General Arrangement with Foundation Loads (in AutoCAD 2010 and PDF)		1E	2	1E
	B03	Cross-section				
C		**Mounting and Erection Drawings**				
	C01	Foundation and Mounting Details (Provided in B02)		1E	4	1E
	C02	Pipe Spools				
	C03	Outline Dimensions, Cut-out and Support Details (Provided in B02)		1E	4	1E
	C04	Match Marked Erection Drawings				
	C05	Motor GA (with Fixing and Shaft Details, and Overall Dimensions)				
D		**Electrical Drawings (in electronic form as appropriate)**				
	D01	Single Line Diagrams				
	D02	Schematic Drawings				
	D03	Terminations / Loop Diagrams		1E	4	1E
	D04	Panel Layout and Fabrication Details				
	D05	Nameplate		1E	4	1E
	D06	Cable Block Drawings				
	D07	Electrical and Instrumentation GAs				
E		**Process and Piping Drawings**				
	E01	Process Flow Sheets				
	E02	Process and Instrument Diagram				
	E03	Pneumatic or Hydraulic Diagram				
	E04	Lubrication Diagram				
F		**Technical Data**				
	F01	Equipment Information Schedule	1E			
	F02	Electrical Performance data sheet		1E	6	
	F03	Equipment List		1E	6	1E
	F04	Vendor's Surface Protection Details				
	F05	Lubrication Data Schedule				
	F06	Material Schedule				
	F07	Bolt Schedule				
	F08	Fan Performance Curves				
	F09	Pump Performance Curves (with Duty Points Marked)				
	F10	Cable Schedule				
	F11	Vendor Catalogue Information	1E			
	F12	Calculations				
	F13	Completed $_{\text{Engineer}}$ Datasheets	1E			
	F14	Motor Datasheets				
	F15	Electrical Equipment List				
	F16	Instrument List				
	F17	I/O List				
	F18	Communications Addressing and Signal Mapping List				

VDDR			Required with Tender	Required after Award		As Built Required
			Copies	Copies	Weeks	Copies
G		**Inspection, Certification and MDR**				
	G01	Inspection and Test Plans	1E	1E	4	
	G02	Inspection and Test Procedures (generic)				
	G03	Quality Plans and Procedures (generic)		1E	4	
	G04	Manufacturer's Data Report (MDR) – Index / Contents		1E	4	
	G05	Manufacturer's Data Report (MDR)		1E	Delivery +2	
	G06	Material Certificates		In MDR		
	G07	Statutory Authority Certificates		In MDR		
	G08	Test Data and Other Certificates		In MDR		
	G09	Welding Procedure / Specification				
	G10	Non-Destructive Testing (NDT) Procedures				
	G11	Material Safety Data Sheets				
	G12	Commissioning Plan and Procedure		1E	Delivery - 4	
	G13	Structural Design Certification				
	G14	Performance Guarantee Statement				
H		**Shipping and Packing Details**				
	H01	Material / Shipping List		1E	Delivery -4	
	H02	Package Type and Quantity (Carton, Case, Crate)		1E	Delivery -4	
	H03	Shipping Packages – Weights and Dimensions		1E	Delivery -4	
	H04	Shipping and Handling Procedure		1E	12	
J		**Manuals**				
	J01	Installation, Operation and Maintenance Manual (Preliminary)		1E	Delivery -4	
	J02	Installation, Operation and Maintenance Manual (Final)		1E	Delivery	
K		**Spare Parts Information**				
	K01	Spare Parts Catalogue and Recommended Spares List (Copy also in Manuals)	1E	1E	8	
	K02	Recommended Spare Parts List with Pricing (Not Included in Manuals)	1E	1E	Prelim 4 Delivery +1	
	K03	Special Tools list	1E	1E	8	
M		**General**				
	M01	Control Philosophy and Logic Diagram				
	M02	Functional Description				